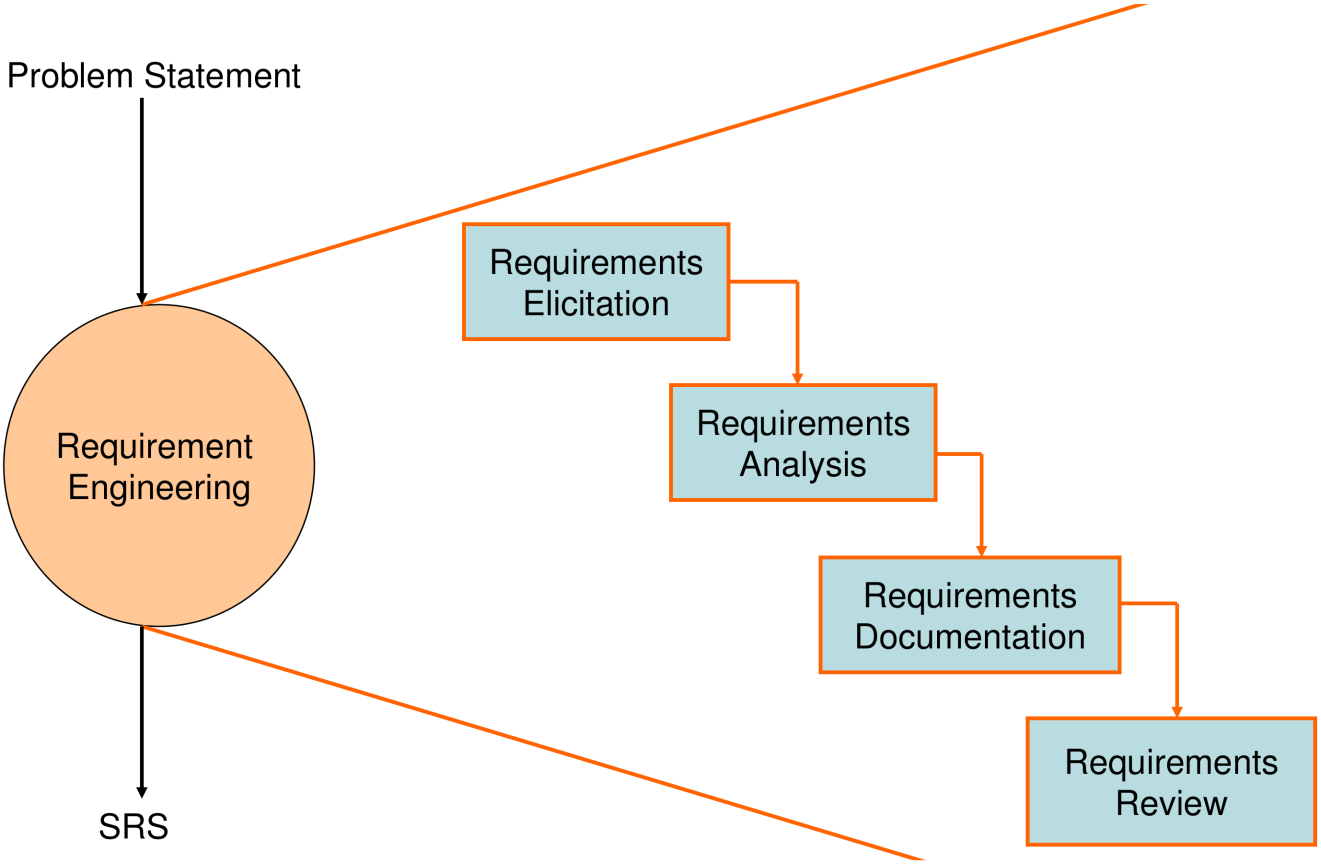
ASE Chapter 3

T4 Architectural Diagrams & Visual Modeling Pack

Section 1: Core Architectural & Process Models (5 Model Units)

Diagram 01. The Four Core Requirements Engineering Activities (Figure 3.1)

Name the input, the four core requirements-engineering activities, and the output shown in the figure.



Requirements engineering process figure

Process shown in the figure:

Input: Problem Statement.

Activities: Requirements Elicitation → Requirements Analysis → Requirements Documentation → Requirements Review.

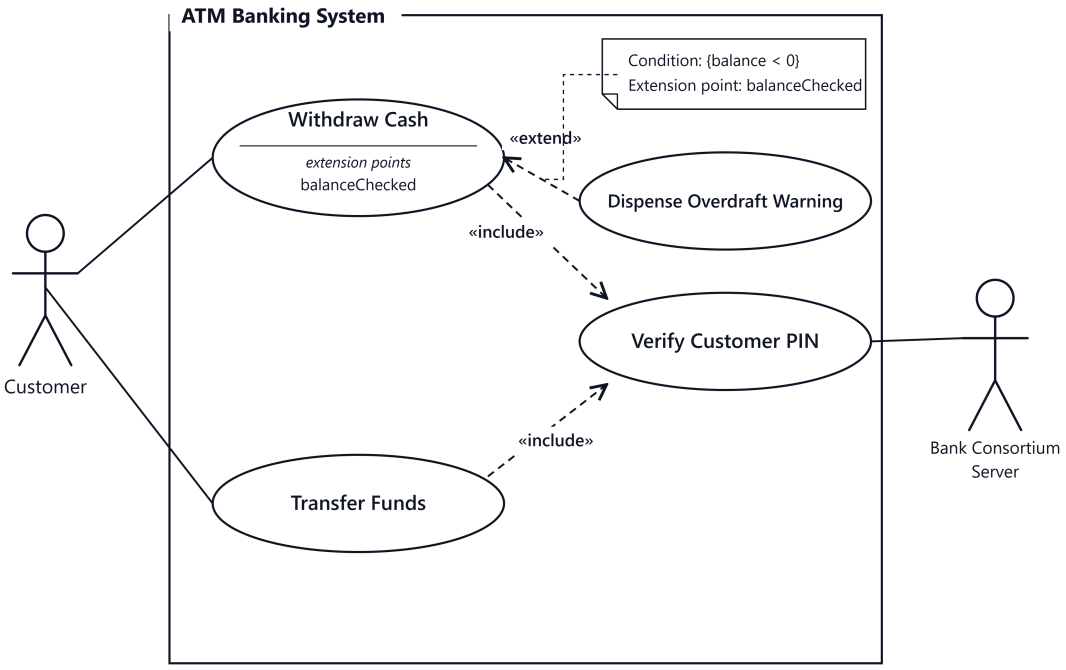
Output: SRS.

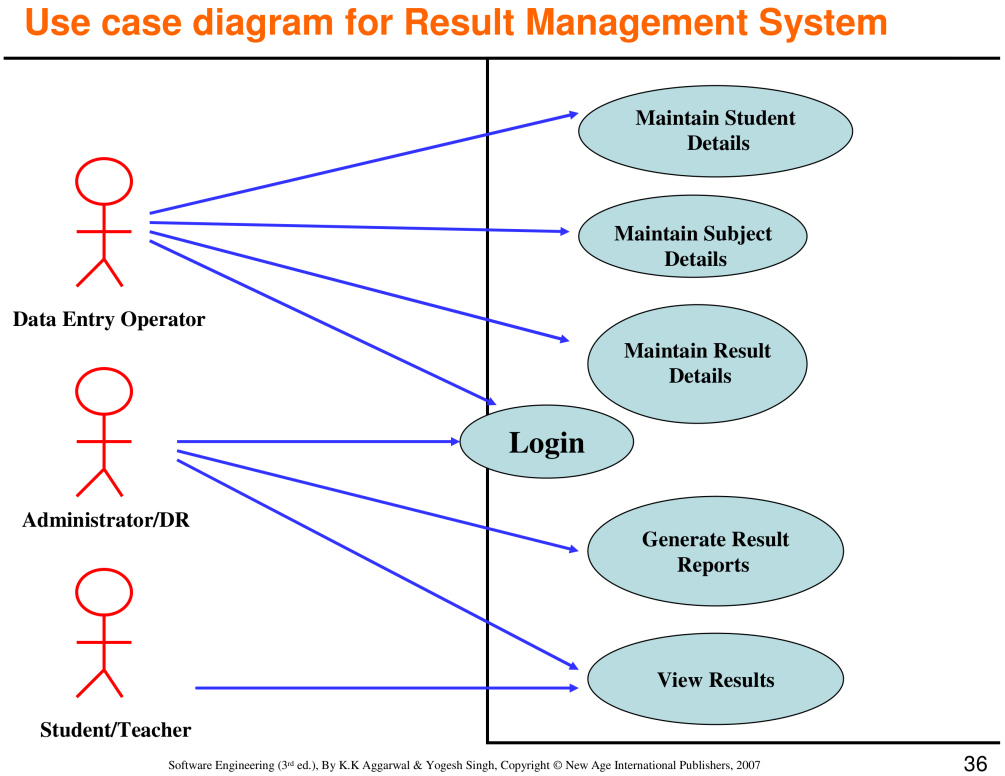
Source: companion Slide 3; K.K. Aggarwal & Yogesh Singh (3rd ed.), Fig. 3.1, p. 41.

> **ملاحظة أكاديمية هامة (Academic Note)**
> الشكل يعرض ترتيب الأنشطة الأربع من دون أسهم رجوع؛ لا يوضّح وحده حلقات مراجعة أو قرار جدوى.

Diagram 02. Use Case Model Architecture: Boundary, Actors, & Stereotypes

Diagram and specify the structural architecture of a UML Use Case Model, illustrating system boundary, external actors, primary use cases, and the defining distinctions between <<include>> and <<extend>> relationships.





Structural & Behavioral Breakdown:

1. **System Boundary:** The rectangular boundary encloses all automated software capabilities, explicitly separating internal functions from external entities.
2. **Primary and Secondary Actors:**
   - Customer (Primary): Initiates interactions to achieve a business goal (Withdraw Cash, Transfer Funds).
   - Bank Consortium Server (Secondary): External backend infrastructure that authenticates and processes transactions.
3. **The <<include>> Stereotype (Mandatory Dependency):**
   - Unconditionally invoked whenever the base use case executes. Both Withdraw Cash and Transfer Funds unconditionally include Verify Customer PIN. The arrow points **from Base Use Case to Included Use Case**.
4. **The <<extend>> Stereotype (Conditional Extension):**
   - Executed only if a specific condition is met at an extension point. Dispense Overdraft Warning extends Withdraw Cash only when the balance drops below zero. The arrow points **from Extending Use Case to Base Use Case**.

Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §3.3.4, pp. 58–61; companion Slide 36 for the system example; OMG UML 2.5.1 UseCases clause for include/extend semantics.

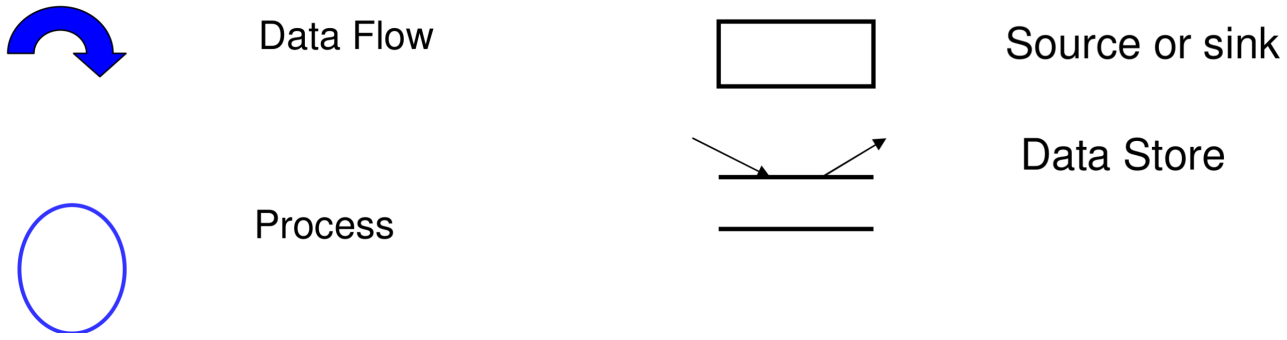
ملاحظة أكاديمية هامة (Academic Note)

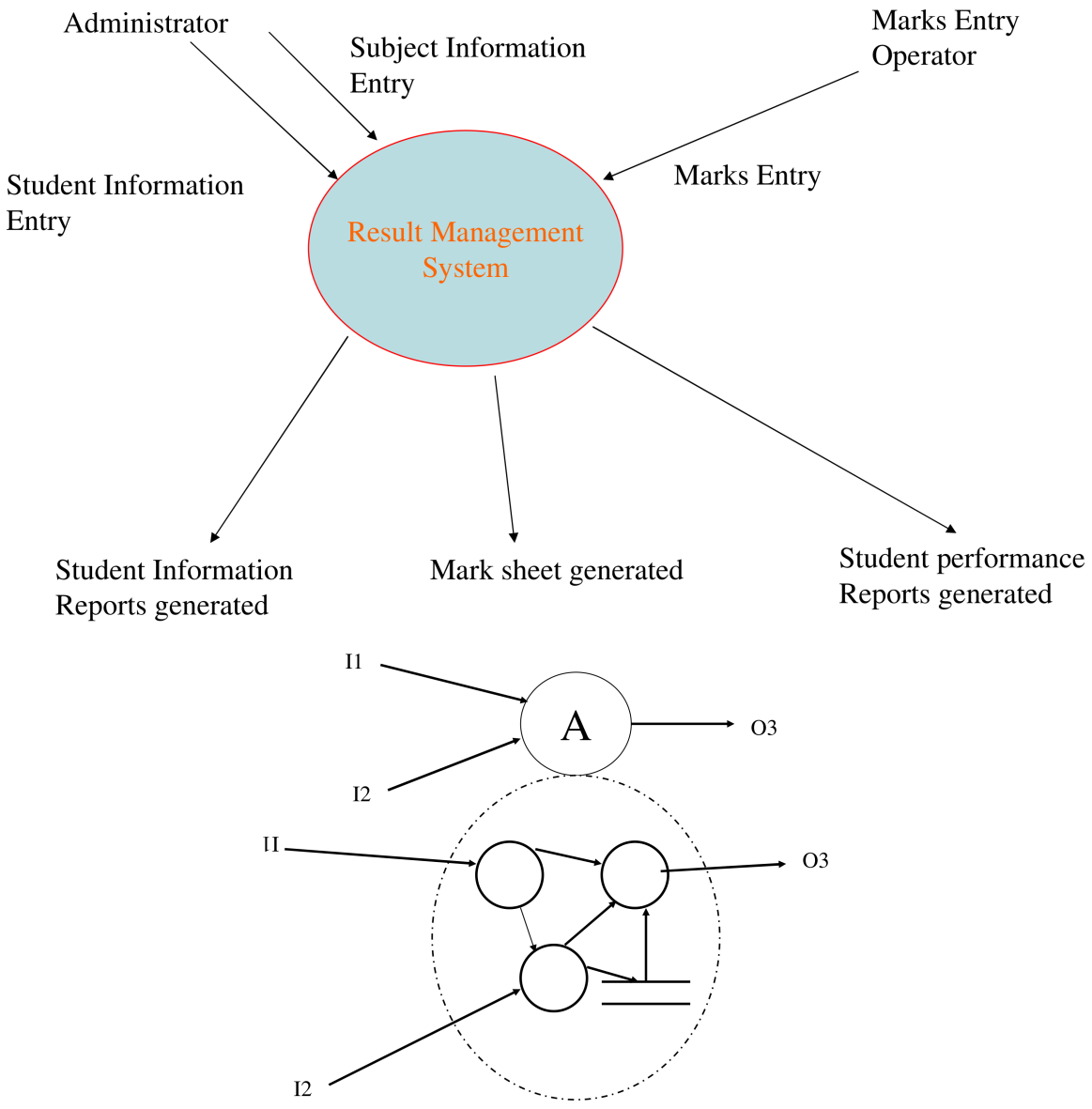
قاعدة السهم في الامتحان:

- في <<include>>: السهم ينطلق من الحالة الأصلية نحو الحالة التابعة (Withdraw Cash → Verify PIN).
- في <<extend>>: السهم ينطلق من الحالة التوسعية الاختيارية عائداً إلى الحالة الأصلية (Overdraft Warning → Withdraw Cash).

Diagram 03. Data Flow Diagram (DFD): Symbols and Separate Examples

Name the four DFD symbols shown, then identify what the publisher examples on Slides 83 and 86 each represent.





DFD decomposition and boundary-flow example

The four symbols:

- **Process:** circle or bubble.
- **Data flow:** directed arrow.
- **Data store:** open or parallel-line symbol.
- **External entity:** rectangle.

Slide 83 shows a Results Management context diagram. Slide 86 shows a separate generic context-to-levelled example using `I1`, `I2`, and `O3`. They are different examples, not a parent diagram and its balanced decomposition. In a genuine decomposition, the child diagram must preserve the parent process's external input and output flows.

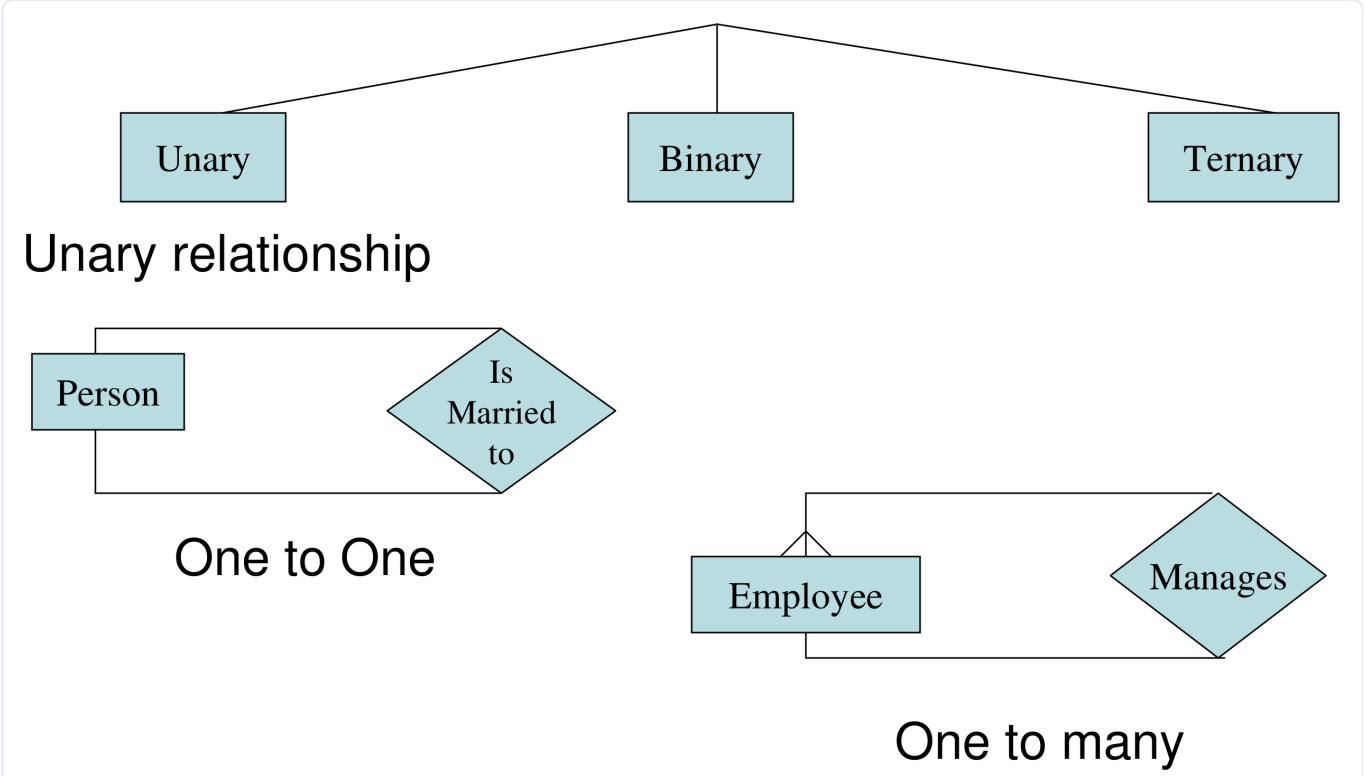
Source: companion Slides 83–86; K.K. Aggarwal & Yogesh Singh (3rd ed.), §3.5.1, pp. 61–65.

ملاحظة أكاديمية هامة (Academic Note)

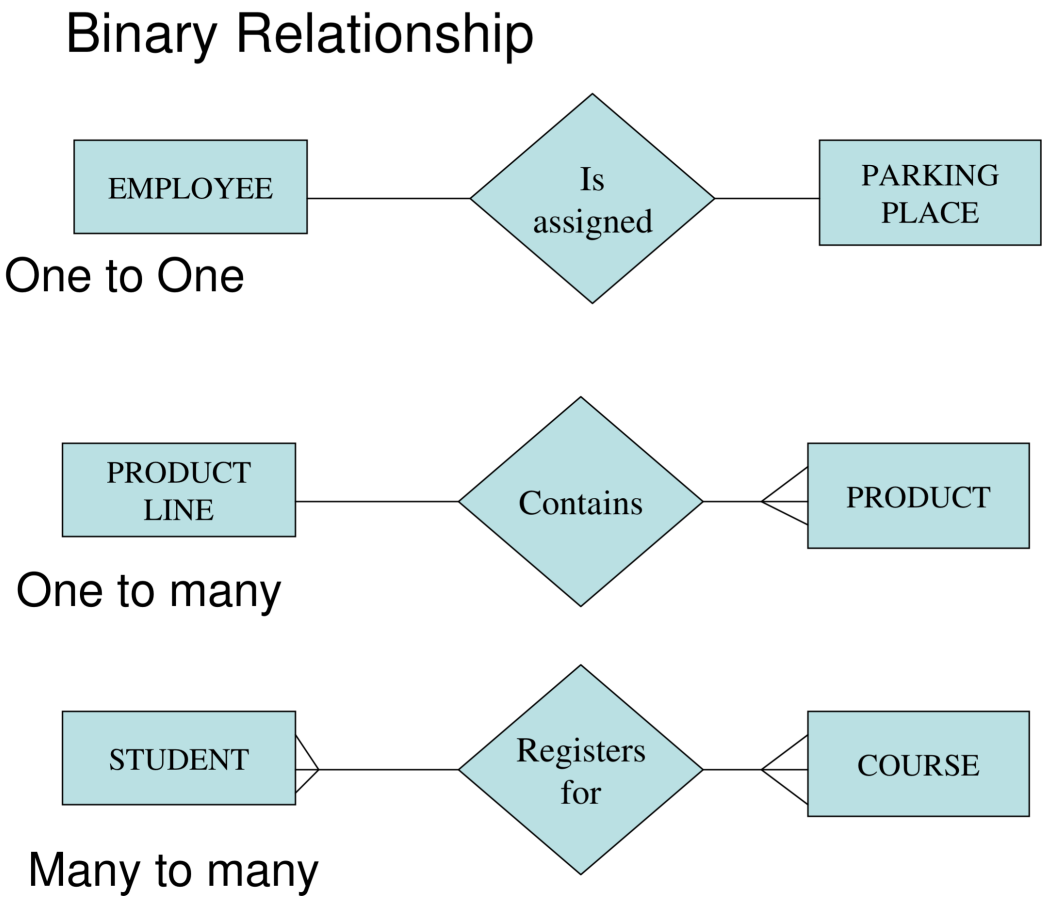
لا نخلط بين مخطط السياق وDFD المستوى الأول في الشريحة 86 وبين مخطط النتائج المستقل في الشريحة 83؛ الشريحتان لا تعرضان زوجاً متوازناً واحداً.

Diagram 04. Entity-Relationship (ER) Architecture & Associative Entity Decomposition

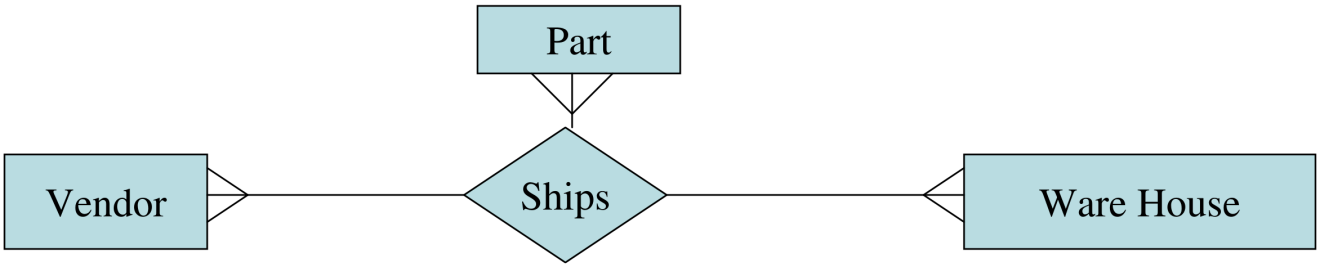
Diagram the core components of an Entity-Relationship Diagram (ERD), illustrate relationship degrees (Unary, Binary, Ternary), and show one relational mapping option for a Many-to-Many (M:N) relationship.



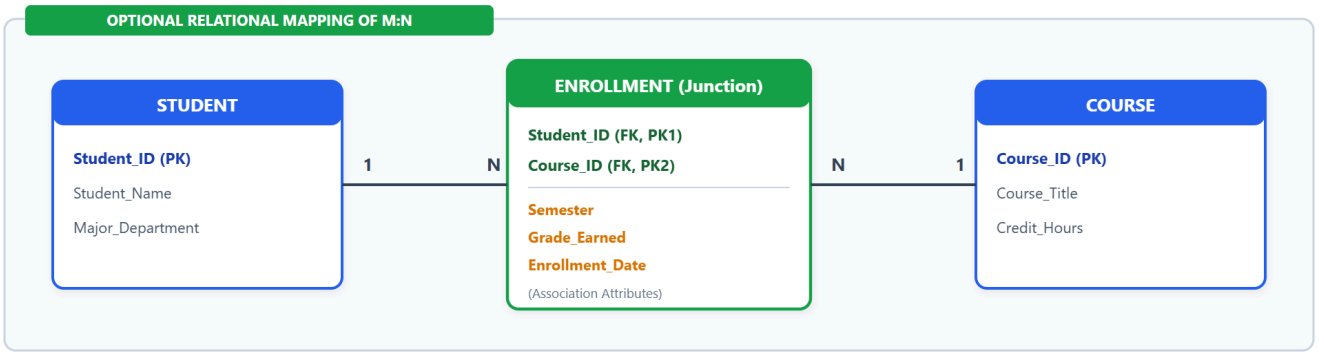
ERD relationship degrees and unary examples



Binary ERD relationship examples



Ternary ERD relationship example



Many-to-many relational mapping illustration

Structural & Relational Walkthrough:

1. **Core ERD Constructs:**
   - **Entity (Rectangle):** Real-world object having independent existence (STUDENT, COURSE).
   - **Attribute (Ellipse):** Properties describing an entity (Student_Name, Credit_Hours).
   - **Relationship (Diamond):** Association between entity instances.
2. **Relationship Degrees:**
   - **Unary (Degree 1):** Associating instances of the same entity type (an EMPLOYEE manages other EMPLOYEEs).
   - **Binary (Degree 2):** Associating instances of two distinct entity types (STUDENT and COURSE).
   - **Ternary (Degree 3):** Simultaneous association among three entity types (DOCTOR, PATIENT, MEDICATION).
3. **A Relational Implementation of M:N:**
   - The conceptual ER model can show a direct Many-to-Many relationship, including attributes of that relationship.
   - When mapping that relationship to relational tables, one common implementation is an **Associative Entity / Junction Table (ENROLLMENT)** with foreign keys such as (Student_ID, Course_ID) and relationship attributes such as Grade, Semester, or Registration_Date. This is a relational mapping option, not a mandatory ER-diagram rule.

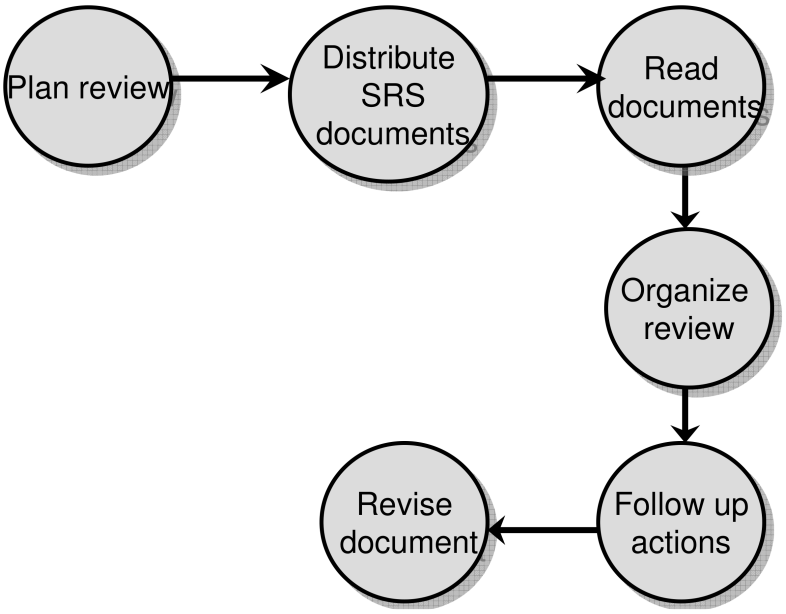
Source: K.K. Aggarwal & Yogesh Singh (3rd ed.), §3.5.2, pp. 66–68; companion Slides 89–95.

ملاحظة أكاديمية هامة (Academic Note)

عند التحويل إلى جداول علائقية: يمكن تمثيل علاقة (M:N) بأحد الخيارات باستخدام كيان رابط (*Associative Entity*) يحمل مفاتيح الكيانين وخصائص العلاقة، مثل الدرجة أو تاريخ التسجيل.

Diagram 05. Requirements Review & Validation Process Workflow (Figure 3.21)

List the six stages of the Requirements Review Process in the order shown.



Six-stage requirements review process

Six stages:

1. Plan review.
2. Distribute SRS documents.
3. Read documents.
4. Organize review.
5. Follow up actions.
6. Revise document.

Source: companion Slide 113; K.K. Aggarwal & Yogesh Singh (3rd ed.), Fig. 3.21, pp. 92–94.

ملاحظة أكاديمية هامة (Academic Note)

الشرائح 114–115 تعرض إجراءات معالجة المشكلات وقائمة التحقق على نحو منفصل؛ الشكل هنا يعرض تسلسل المراحل الست فقط.